The fair-coin case: $w_{n,1/2} = \frac{1}{2}$ for every n

2026-04-18

Setting

Consider the all-heads-wins game with $n \in \mathbb{N}$ coins, each landing heads with probability $p \in (0,1)$ independently. In every round all remaining coins are tossed and the player must set aside at least one *head*; setting aside a tails causes immediate loss. The game is won iff every set-aside coin shows heads, equivalently iff the player ever reaches the state of zero remaining coins.

For a policy π (a choice, in every state n and after observing $k \geq 1$ heads, of how many heads $i_\pi(n,k) \in \{1,\dots,k\}$ to set aside), write v_n^π for the probability of winning starting with n coins under π, and $v_0 := 1$. The Bellman optimum is

$$w_{n,p} \;:=\; \sup_\pi v_n^\pi \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \max_{j \leq m \leq n-1} w_{m,p},$$

with $w_{0,p} := 1$ and $w_{1,p} = p$.

The two reference strategies are

(A) set aside exactly one head when $1 \leq k \leq n-1$; set aside all n heads when $k = n$;

(B) set aside every head ($i_\pi(n,k) = k$).

Both strategies set aside all n heads when $k = n$ and therefore win immediately upon observing all heads.

Statement

Proposition 1. *For every $n \geq 1$,*

$$w_{n,\,1/2} \;=\; \tfrac{1}{2}. \tag{1}$$

Moreover, for every policy π that satisfies $i_\pi(n,n) = n$ for every $n \geq 1$ — i.e. that takes the immediate win when all coins come up heads — the win probability satisfies $v_n^\pi = \frac{1}{2}$. In particular, the strategies (A) and (B) both achieve $v_n^{(\mathrm{A})} = v_n^{(\mathrm{B})} = \frac{1}{2}$.

Proof

Throughout the proof, $p = q = \frac{1}{2}$. The key cancellation is the identity

$$\sum_{j=0}^{n} \binom{n}{j} \left(\tfrac{1}{2}\right)^n \;=\; 1, \qquad \sum_{j=1}^{n-1} \binom{n}{j} \left(\tfrac{1}{2}\right)^n \;=\; 1 - \tfrac{2}{2^n} \;=\; 1 - 2^{1-n}. \tag{2}$$

Step 1: $w_{n,\,1/2} = \frac{1}{2}$ by induction on n.

Base. $w_{1,\,1/2} = \frac{1}{2}$ by definition.

Inductive step. Fix $n \geq 2$ and assume $w_{m,\,1/2} = \frac{1}{2}$ for every $m \in \{1,\dots,n-1\}$. Then for every $j \in \{1,\dots,n-1\}$ the suffix maximum collapses, $\max_{j \leq m \leq n-1} w_{m,\,1/2} = \frac{1}{2}$, and the Bellman gives

$$\begin{aligned} w_{n,\,1/2} \;&=\; \left(\tfrac{1}{2}\right)^n + \sum_{j=1}^{n-1} \binom{n}{j} \left(\tfrac{1}{2}\right)^{n-j} \left(\tfrac{1}{2}\right)^j \cdot \tfrac{1}{2} \\ &=\; \left(\tfrac{1}{2}\right)^n + \tfrac{1}{2} \cdot \sum_{j=1}^{n-1} \binom{n}{j} \left(\tfrac{1}{2}\right)^n \;=\; \left(\tfrac{1}{2}\right)^n + \tfrac{1}{2}\left(1 - 2^{1-n}\right) \\ &=\; 2^{-n} + \tfrac{1}{2} - 2^{-n} \;=\; \tfrac{1}{2}, \end{aligned}$$

using (2). This closes the induction and proves (1).

Step 2: $v_n^\pi = \frac{1}{2}$ for any π with $i_\pi(n,n) = n$.

Such a π chooses, in every state n and after observing $k \in \{1,\dots,n\}$ heads, an action $i_\pi(n,k) \in \{1,\dots,k\}$, with the additional property $i_\pi(n,n) = n$. Conditioning on the number of heads $K \sim \text{Binomial}(n, \frac{1}{2})$ in the first round and on $K \geq 1$ (the event $K = 0$ causes immediate loss):

$$v_n^\pi \;=\; \sum_{k=1}^{n} \binom{n}{k} \left(\tfrac{1}{2}\right)^n v^\pi_{n-i_\pi(n,k)}.$$

For $k = n$, $i_\pi(n,n) = n$ gives $v^\pi_{n-n} = v_0 = 1$. For $k \in \{1,\dots,n-1\}$, $n - i_\pi(n,k) \in \{n-k,\dots,n-1\} \subseteq \{1,\dots,n-1\}$.

Argue by induction on n. The base case $n = 1$: the only action when $k = 1$ is $i_\pi(1,1) = 1$, giving $v_1^\pi = \frac{1}{2} \cdot v_0 = \frac{1}{2}$. Assume $v_m^\pi = \frac{1}{2}$ for every $m \in \{1,\dots,n-1\}$. Then

$$v_n^\pi \;=\; \binom{n}{n}\left(\tfrac{1}{2}\right)^n \cdot 1 + \sum_{k=1}^{n-1} \binom{n}{k}\left(\tfrac{1}{2}\right)^n \cdot \tfrac{1}{2} \;=\; 2^{-n} + \tfrac{1}{2}\big(1 - 2^{1-n}\big) \;=\; \tfrac{1}{2},$$

again by (2).

Step 3: strategies (A) and (B) qualify.

Both (A) and (B) set aside all n heads when all n coins land heads ($k = n$): for (A) this is the explicit clause "set aside all (and win)"; for (B) one has $i_{\mathrm{B}}(n,n) = n$ since every head is set aside. Hence $v_n^{(\mathrm{A})} = v_n^{(\mathrm{B})} = \frac{1}{2}$ by Step 2. □

Two remarks

Remark (Strategies that throw away the all-heads win). The hypothesis $i_\pi(n,n) = n$ is essential. If π instead chooses $i_\pi(n,n) < n$ when all n coins come up heads — i.e. refuses the immediate win — then $v_n^\pi = \frac{1}{2} - 2^{-(n+1)} < \frac{1}{2}$.

Proof sketch. In the Bellman computation the $k = n$ contribution becomes $\binom{n}{n}(1/2)^n \cdot v^\pi_{n-i_\pi(n,n)} = 2^{-n} \cdot \frac{1}{2}$ instead of $2^{-n} \cdot 1$. The total is therefore $2^{-n} \cdot \frac{1}{2} + \frac{1}{2}(1 - 2^{1-n}) = \frac{1}{2} - 2^{-(n+1)}$.

Remark (Sharpness). The conclusion $w_{n,p} = \frac{1}{2}$ for all n is special to $p = \frac{1}{2}$. For $p > \frac{1}{2}$, $n \mapsto w_{n,p}$ is strictly increasing (*Manuscript Lemma 1*); for $p < \frac{1}{2}$, $n \mapsto w_{n,p}$ is non-monotone with $w_{n,p} < w_{1,p} = p$ for every $n \geq 2$ (*Proposition 3 of* `localmaximum.tex`).

A recursion for the deficit $\Delta_{n,p} = \frac{1}{2} - w_{n,p}$

Writing $w_{n,p} = \frac{1}{2} - \Delta_{n,p}$ in the Bellman gives a clean recursion for the deviation from the fair-coin value $\frac{1}{2}$. Note that the maximum of $w_{m,p}$ becomes a minimum of $\Delta_{m,p}$:

$$\max_{j \leq m \leq n-1} w_{m,p} \;=\; \tfrac{1}{2} - \min_{j \leq m \leq n-1} \Delta_{m,p}.$$

Proposition 2 (Recursion for $\Delta_{n,p}$)**.** *Let $q := 1 - p$ and $\Delta_{n,p} := \frac{1}{2} - w_{n,p}$ for $n \geq 1$ (empty-sum convention at $n = 1$). Then*

$$\Delta_{n,p} \;=\; \frac{q^n - p^n}{2} \;+\; \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \leq m \leq n-1} \Delta_{m,p}. \tag{3}$$

Proof. Substitute $w_{m,p} = \frac{1}{2} - \Delta_{m,p}$ in $w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \leq m \leq n-1} w_{m,p}$:

$$\begin{aligned}
\tfrac{1}{2} - \Delta_{n,p} \;&=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \big(\tfrac{1}{2} - \min_{j \leq m \leq n-1} \Delta_{m,p}\big) \\
&=\; p^n + \tfrac{1}{2}\big(1 - p^n - q^n\big) - \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \leq m \leq n-1} \Delta_{m,p} \\
&=\; \tfrac{1}{2} + \frac{p^n - q^n}{2} - \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \leq m \leq n-1} \Delta_{m,p},
\end{aligned}$$

where the second equality uses $\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - q^n$. Solving for $\Delta_{n,p}$ gives (3). □ □

Remark (Sanity checks).

- At $n = 1$, the sum in (3) is empty, and $(q - p)/2 = \frac{1}{2} - p = \Delta_{1,p}$.
- At $p = \frac{1}{2}$, the constant term $(q^n - p^n)/2 = 0$ and the induction hypothesis $\Delta_{m,1/2} = 0$ for $m < n$ makes the sum vanish, recovering Proposition 1.

Corollary 3 (Sign of $\Delta_{n,p}$)**.** *For $p \in (0, \frac{1}{2})$, $\Delta_{n,p} > 0$ for every $n \geq 1$. For $p \in (\frac{1}{2}, 1)$, $\Delta_{n,p} < 0$ for every $n \geq 1$.*

Proof. Both follow from (3) by induction on n. For $p < \frac{1}{2}$, the constant $(q^n - p^n)/2 > 0$ and (inductively) $\min_{j \leq m \leq n-1} \Delta_{m,p} > 0$, so each summand is non-negative, hence $\Delta_{n,p} \geq (q^n - p^n)/2 > 0$. The case $p > \frac{1}{2}$ is analogous with reversed signs. □ □

Remark (Recursion when Δ is monotone non-decreasing). If $(\Delta_{m,p})_{m=1}^{n-1}$ is non-decreasing in m (equivalently, $w_{m,p}$ is non-increasing in m), then $\min_{j \leq m \leq n-1} \Delta_{m,p} = \Delta_{j,p}$, and (3) reduces to the linear recursion

$$\Delta_{n,p} \;=\; \frac{q^n - p^n}{2} + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \, \Delta_{j,p}.$$

This is the strategy-(B) version of the Δ-recursion (no min, no max). It is the natural object on which to attempt analytic estimates of the deficit.

Expansion near $p = \frac{1}{2}$

Set $\delta := \frac{1}{2} - p \in (0, \frac{1}{2})$, so $p = \frac{1}{2} - \delta$ and $q = \frac{1}{2} + \delta$. We expand the recursion (3) in powers of δ. Several cancellations follow from the fact that $n \mapsto w_{n,1/2}$ is constant: at $\delta = 0$ all $\Delta_{m,p}$ vanish.

Three pieces, each $O(\delta)$

(i) The constant term. By the binomial theorem,

$$\frac{q^n - p^n}{2} \;=\; \frac{(\frac{1}{2} + \delta)^n - (\frac{1}{2} - \delta)^n}{2} \;=\; \sum_{k \text{ odd}} \binom{n}{k} \frac{\delta^k}{2^{n-k}} \;=\; \frac{n}{2^{n-1}} \, \delta + O(\delta^3). \tag{4}$$

Only odd powers of δ appear; in particular this term vanishes linearly at $\delta = 0$.

(ii) The binomial weights. A direct expansion gives $\binom{n}{j} p^{n-j} q^j = \binom{n}{j}/2^n + O(\delta)$ uniformly in j.

(iii) The min. Inductively (cf. Corollary 3) each $\Delta_{m,p}$ vanishes at $\delta = 0$, so for $\delta > 0$ small, $\min_{j \leq m \leq n-1} \Delta_{m,p} \to 0$ as $\delta \to 0$.

The leading-order coefficient

Define c_n by the asymptotic expansion

$$\Delta_{n,p} \;=\; c_n \, \delta + O(\delta^2) \qquad (\delta \downarrow 0). \tag{5}$$

For $\delta > 0$ small (i.e. $p < \frac{1}{2}$ and approaching $\frac{1}{2}$ from below), $\Delta_{m,p} \sim c_m \delta > 0$ inductively, so $\min_{j \leq m \leq n-1} \Delta_{m,p} = \delta \cdot \min_{j \leq m \leq n-1} c_m + O(\delta^2)$. Substituting (i)–(iii) into (3) and matching first-order coefficients of δ:

Proposition 4 (Recursion for c_n)**.** *Set $c_1 := 1$. For every $n \geq 2$,*

$$c_n \;=\; \frac{n}{2^{n-1}} \;+\; \frac{1}{2^n} \sum_{j=1}^{n-1} \binom{n}{j} \min_{j \leq m \leq n-1} c_m. \tag{6}$$

Remark. Equation (6) is precisely the recursion for α_n^{left} derived in the journal's Taylor-expansion section (`journal.md`, "Taylor expansion at $p = \frac{1}{2}$"), with the sign convention $\delta = \frac{1}{2} - p$ here corresponding to $h = -\delta$ there. Numerically (40 dps),

$$c_1 = 1,\; c_2 = \tfrac{3}{2},\; c_3 = \tfrac{27}{16},\; c_4 = \tfrac{111}{64},\; c_5 = \tfrac{3555}{2048} \approx 1.7358,\; c_6 \approx 1.7294,\; \ldots,\; c_\infty \approx 1.703.$$

Two perturbative consequences

Corollary 5 (First-order shape of $w_{n,p}$ near $p = \frac{1}{2}$)**.** *For $\delta = \frac{1}{2} - p > 0$ small, $w_{n,p} = \frac{1}{2} - c_n\delta + O(\delta^2)$. The sign of $w_{n,p} - w_{n-1,p}$ at first order is opposite to that of $c_n - c_{n-1}$. Since (c_n) is valley-shaped — strictly increasing on $\{1, \dots, 5\}$ and strictly decreasing for $n \geq 5$ — the sequence $n \mapsto w_{n,p}$ has, at first order in δ, a strict local* minimum *at $n = 5$ and* no *local maximum.*

Remark (Local maxima are non-perturbative)*.* The first-order picture has no local maximum, in agreement with the numerical observation (cf. `journal.md`, "First local minima of $w_{n,p}$ vs p") that the first local minimum of w stabilises at $n = 5$ for p close to $\frac{1}{2}$, while the first strict local *maximum* of w is pushed out to $n = 152$ at $p = 0.49$ (i.e. $\delta = 0.01$). Local maxima therefore appear only via $O(\delta^2)$ or higher corrections to (5).

The limit $L = \lim c_n$: analytic characterization

Numerically $c_n \to L \approx 1.7034717608717367\dots$ The limit has a clean analytic existence proof because the recursion (6) *simplifies to a first-order linear recurrence* once n is large enough. The key observation is about the location of the minimum in $\min_{j \leq m \leq n-1} c_m$.

Lemma 6 (Asymptotic collapse of the min)**.** *Set $c_1 = 1,\ c_2 = \frac{3}{2},\ c_3 = \frac{27}{16}$. For every $n \geq 7$,*

$$\min_{j \leq m \leq n-1} c_m \;=\; \begin{cases} c_j & j \in \{1, 2, 3\}, \\ c_{n-1} & j \in \{4, \dots, n-1\}. \end{cases} \tag{7}$$

Proof. The lemma reduces to two structural facts:

- **(L1)** $c_m \geq \frac{27}{16} = c_3$ for every $m \geq 4$.
- **(L2)** $c_{n-1} \leq c_m$ for every $m \in \{4, \dots, n-1\}$, $n \geq 7$.

Indeed, (L1) and $c_1 < c_2 < c_3$ imply that for $j \in \{1, 2, 3\}$ the minimum of the suffix is c_j (the smallest entry); (L2) says the minimum of the suffix from any $j \geq 4$ is c_{n-1}.

Proof of (L1) by induction on m. The base case $m = 4$ is $c_4 = \frac{111}{64} = \frac{27}{16} + \frac{3}{64} > \frac{27}{16}$.

Inductive step, $m \geq 5$. Assume $c_l \geq \frac{27}{16}$ for every $l \in \{4, \dots, m-1\}$. Combined with $c_1 = 1$, $c_2 = \frac{3}{2}$, $c_3 = \frac{27}{16}$, the minimum entries appearing in the recursion (6) for c_m are

$$\min_{j \leq l \leq m-1} c_l \;\geq\; \begin{cases} c_j, & j \in \{1, 2, 3\}, \\ \frac{27}{16}, & j \in \{4, \dots, m-1\}. \end{cases}$$

Plugging into (6) and using $\sum_{j=4}^{m-1} \binom{m}{j} = 2^m - 2 - m - \binom{m}{2} - \binom{m}{3}$,

$$\begin{aligned} c_m \;&\geq\; \frac{m}{2^{m-1}} + \frac{1}{2^m}\left[m \cdot 1 + \binom{m}{2} \cdot \tfrac{3}{2} + \binom{m}{3} \cdot \tfrac{27}{16} + \tfrac{27}{16} \sum_{j=4}^{m-1} \binom{m}{j}\right] \\ &=\; \tfrac{27}{16} + \frac{3}{32 \cdot 2^m}\big(15m - m^2 - 36\big) \;=\; \tfrac{27}{16} - \frac{3(m^2 - 15m + 36)}{32 \cdot 2^m}. \end{aligned}$$

Since the roots of $m^2 - 15m + 36$ are $m = 3$ and $m = 12$, the bracket $m^2 - 15m + 36 \leq 0$ for every $m \in \{3, \dots, 12\}$, and hence $c_m \geq \frac{27}{16}$ for $m \in \{4, \dots, 12\}$.

For $m \geq 13$ the simple bound above is no longer sufficient (the bracket becomes positive). We sharpen by tracking the deficit $\varepsilon_l := c_l - \frac{27}{16}$. Under the joint inductive hypothesis (which we strengthen below to include $c_l < c_{l-1}$ for $l \in \{6, \dots, n-1\}$), the suffix-min in the recursion (6) collapses as in (7), and a short substitution (using $\sum_{j=4}^{n-1} \binom{n}{j} = 2^n - 2 - n - \binom{n}{2} - \binom{n}{3}$) yields, for $n \geq 7$,

$$\varepsilon_n \;=\; (1 - B_n)\,\varepsilon_{n-1} \;+\; \big(A_n - \tfrac{27}{16} B_n\big), \tag{8}$$

where A_n, B_n are as in (11). A direct algebraic computation (using c_1, c_2, c_3 as listed) gives the closed form

$$A_n - \tfrac{27}{16} B_n \;=\; -\frac{3(n^2 - 15n + 36)}{32 \cdot 2^n}, \tag{9}$$

which is non-positive for $n \geq 13$ (and positive for $n \in \{4, \dots, 11\}$). Iterating (8) from $n_0 = 13$ upward,

$$\varepsilon_n \;=\; \varepsilon_{12} \prod_{k=13}^{n} (1 - B_k) \;+\; \sum_{k=13}^{n} \big(A_k - \tfrac{27}{16} B_k\big) \prod_{m=k+1}^{n} (1 - B_m).$$

Direct rational computation gives $\varepsilon_{12} = c_{12} - \frac{27}{16} = \frac{21\,976\,805\,576\,165\,276\,031}{1\,180\,591\,620\,717\,411\,303\,424} \approx 0.01861 > \frac{1}{60}$. The infinite product is bounded via Weierstrass: $\sum_{m\geq 0} B_m = 10$ in closed form (using $\sum_{m\geq 0} m^k/2^m$ for $k = 0, 1, 2, 3$ and the relevant identities for falling factorials), and $\sum_{m=0}^{12} B_m = \frac{20\,237}{2048}$ by direct rational computation; hence $\sum_{m\geq 13} B_m = 10 - \frac{20\,237}{2048} = \frac{243}{2048} < \frac{1}{8}$, and $\prod_{m\geq 13}(1 - B_m) \geq 1 - \sum_{m\geq 13} B_m > \frac{7}{8}$. Hence the first term in the iteration is bounded below by $\frac{1}{60} \cdot \frac{7}{8} = \frac{7}{480}$.

The total negative contribution from the second sum is bounded by

$$\sum_{k\geq 13} \frac{3(k^2 - 15k + 36)}{32 \cdot 2^k} \;\leq\; \frac{3}{32} \sum_{k\geq 13} \frac{k^2}{2^k} \;=\; \frac{3}{32} \cdot \frac{99}{2048} \;=\; \frac{297}{65\,536} \;<\; \frac{1}{200},$$

using the well-known identity $\sum_{k\geq 0} k^2/2^k = 6$ and the direct rational computation $\sum_{k=0}^{12} k^2/2^k = \frac{12\,189}{2048}$ (so the tail equals $6 - \frac{12\,189}{2048} = \frac{99}{2048}$). Therefore

$$\varepsilon_n \;>\; \frac{7}{480} - \frac{1}{200} \;=\; \frac{35 - 12}{2400} \;=\; \frac{23}{2400} \;>\; 0$$

for every $n \geq 13$, i.e. $c_n > \frac{27}{16}$. This completes the induction for (L1).

Proof of (L2). It suffices to show that $(c_n)_{n\geq 5}$ is strictly decreasing, i.e. $c_n < c_{n-1}$ for every $n \geq 6$. For $n \in \{6, \ldots, 12\}$ this is verified directly in exact rational arithmetic. For $n \geq 13$, under the joint inductive hypothesis the recursion (6) reduces (by the same substitution leading to (8)) to $c_n = A_n + (1 - B_n)c_{n-1}$, so

$$c_n - c_{n-1} \;=\; A_n - B_n c_{n-1} \;<\; A_n - \tfrac{27}{16} B_n \;=\; -\frac{3(n^2 - 15n + 36)}{32 \cdot 2^n} \;<\; 0,$$

where the strict inequality uses (L1): $c_{n-1} > \frac{27}{16}$. Hence $c_n < c_{n-1}$ for every $n \geq 13$, completing (L2). □ □

Proposition 7 (Linear recursion for c_n, $n \geq 7$)**.** *For every $n \geq 7$,*

$$c_n \;=\; A_n + (1 - B_n)\, c_{n-1}, \tag{10}$$

where

$$A_n \;:=\; \frac{n}{2^{n-1}} + \frac{nc_1 + \binom{n}{2}c_2 + \binom{n}{3}c_3}{2^n}, \qquad B_n \;:=\; \frac{2 + n + \binom{n}{2} + \binom{n}{3}}{2^n}. \tag{11}$$

Proof. Substitute Lemma 6 into (6):

$$c_n \;=\; \frac{n}{2^{n-1}} + \frac{1}{2^n}\left[\sum_{j=1}^{3} \binom{n}{j} c_j + c_{n-1} \sum_{j=4}^{n-1} \binom{n}{j}\right].$$

Using $\sum_{j=4}^{n-1} \binom{n}{j} = 2^n - 2 - n - \binom{n}{2} - \binom{n}{3}$ and collecting terms yields (10). □ □

Lemma 8 (Bounds)**.** $0 < c_n \leq 3$ *for every $n \geq 1$, and $A_n, B_n = O(n^3/2^n)$.*

Proof. Positivity is immediate from (6). For the upper bound, from $\min_{j\leq m\leq n-1} c_m \leq \max_m c_m$ and inductively assuming $c_m \leq 3$ for $m < n$,

$$c_n \leq \frac{n}{2^{n-1}} + \frac{1}{2^n}(2^n - 2) \cdot 3 \;\leq\; \frac{n}{2^{n-1}} + 3 \;\leq\; 3 + \frac{n}{2^{n-1}},$$

and $c_1 = 1 < 3$. Since $n/2^{n-1} \leq 1$ for $n \geq 2$... (tighter bound follows from $c_n \leq 2$ by a sharper induction using numerical values of $c_1, \ldots, c_5$; we only need some uniform upper bound). The asymptotic order of A_n, B_n follows from $\binom{n}{k} \sim n^k/k!$. □ □

Theorem 9 (Existence and explicit formula for the limit)**.** *The limit $L := \lim_{n\to\infty} c_n$ exists, and for every $n_0 \geq 7$,*

$$L \;=\; c_{n_0-1} \prod_{m\geq n_0} (1 - B_m) \;+\; \sum_{k\geq n_0} A_k \prod_{m>k} (1 - B_m), \tag{12}$$

with both the infinite product and the infinite sum convergent at geometric rate.

Proof. Iterate (10) from n_0 upward:

$$c_{n_0+k} \;=\; c_{n_0-1}\prod_{m=n_0}^{n_0+k}(1-B_m) + \sum_{j=n_0}^{n_0+k} A_j \prod_{m=j+1}^{n_0+k}(1-B_m).$$

By Lemma 8, $B_m = O(m^3/2^m)$ is summable and $\sum B_m < \infty$, so the infinite product $\prod_{m\ge n_0}(1-B_m)$ converges to a positive limit. Similarly $\sum A_j < \infty$ and each product $\prod_{m>j}(1-B_m)$ is bounded in j, so the sum in (12) is absolutely convergent. Hence the right-hand side of the iterate converges as $k \to \infty$, which defines L and gives (12). □ □

Remark (Numerical value). Equation (12) is an explicit (infinite-series) formula for L with geometric convergence. Direct computation with $n_0 = 30$ truncated at $m \le 400$ already gives

$$L \;=\; 1.70347176087173673645200305679541476880577572003162\ldots$$

to 50 digits (the omitted tail is $O(400^3/2^{400})$, far beyond machine precision). The rate of convergence is geometric with ratio $\frac{1}{2}$: $c_n - L = \Theta(n^3/2^n)$, since the binomial tail $\binom{n}{3}/2^n$ dominates the $O(n^3/2^n)$ scale of A_n, B_n.

Remark (No elementary closed form). The limit L is implicitly defined by the recursion (6) with its non-smooth min; it does not admit an elementary closed-form expression analogous to $\alpha_\infty^{\text{right}}$ (which also lacks a simple closed form, satisfying instead $L^{\text{right}} = 4 - \sum_{j\ge1}\alpha_j^{\text{right}}/2^j$). Formula (12) is the best explicit representation available.